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TRAUMATIC HAEMORRHAGE

SIGNIFICANT EXTERNAL HAEMORRHAGE SHOULD BE CONTROLLED FIRST. INTERNAL HAEMORRHAGE MAY BE INCOMPRESSIBLE SO DO NOT DELAY TRANSPORT TO DAMAGE CONTROL SURGERY (DCS)

EXTERNAL HAEMORRHAGE CONTROL

LIMBS = TOURNIQUET (TQ)

CUF	PLACE TQ HIGH AND TIGHT
PERMISSIVE	TQ LOW AS POSSIBLE ON LIMB

WOUND SHOULD BE ASSESSED AND TQ DOWNGRADED AS SOON AS POSSIBLE IF ENVIRONMENT ALLOWS.

JUNCTIONAL (ARMPIT or GROIN) = PACK

DIRECT PRESSUE > PACK WITH CELOX/GAUZE > SECURE WOUND PACKING WITH ECB

DO NOT USE CELOX IN HEAD INJURIES WITH BRAIN MATTER SHOWING OR TORSO WOUNDS

NON-COMPRESSIBLE OR INTERNAL HAEMORRHAGE

TORSO (CHEST or ABDOMINAL)

CONVEY TO DCS FACILITY (ROLE 2 or 3/MTC)
CAUTIOUS FLUID/BLOOD RESUS
CONSIDER PERMISSIVE HYPOTENSION (see below)

PELVIC FRACTURE

MOI AND: PAIN or LEGS EXTERNALLY ROTATED or ASSYMETRICAL PELVIS or SHOCKED PATIENT

PELVIC BINDER AND CONSIDER FLUID RESUS

MONITOR FOR SIGNS OF SHOCK (HATEFUL 8)

WORSENING TACHYCARDIA	NO PALPABLE RADIAL PULSE or SBP <90mmHg	FALLING ETCO$_2$ LEVELS	PALENESS or GREYNESS
REDUCED CONSCIOUS LEVELS	VENOUS COLLAPSE	“AIR HUNGER”	CLAMMY/SWEATING

MANAGEMENT

GAIN IV/IO ACCESS
ADMINISTER 1g TRANEXAMIC ACID (TXA)
CONSIDER 250ml FLUID BOLUSES FOR SHOCKED PATIENTS or BLOOD PRODUCTS IF AVAILABLE

PREHOSPITAL BLOOD TRANSFUSION (PHBT)

AIM = AVOID HYPOPERFUSION TO VITAL ORGANS and ACUTE TRAUMATIC COAGULOPATHY
ADMINISTER PACKED RED BLOOD CELLS (PRBC) AND LYOPLAS/FFP IN 1:1 RATIO
ADMINISTER CALCIUM CHLORIDE 10% 10ml AFTER EACH BAG OF PRBC
REASSESS PERFUSION STATUS AFTER EACH BLOOD PRODUCT
ASSESS FOR TRANSFUSION REACTIONS EVERY 15 MINUTES

SEVERE TRANSFUSION REACTION	STOP TRANSFUSION IF SEVERE ALLERGIC REACTION or ANAPHYLAXIS OCCURS THEN MANAGE IN ACCORDANCE WITH ADVANCED LIFE SUPPORT GUIDELINES
MILD/MODER TRANSFUSION REACTION	CONSIDER SLOWING TRANSFUSION AND ADMINISTER IN A SEPARATE IV ACCESS CHLORPHENAMINE 10mg and HYDROCORTISONE 200mg, SALBUTAMOL 5mg NEB

PERMISSIVE HYPOTENSION GUIDANCE

INDICATIONS	CONSIDER IN PATIENTS WITH NON-COMPRESSIBLE TORSO HAEMORRHAGE, ESPECIALLY WITH PENETRATING THORACIC TRAUMA WITHIN 1 HOUR OF INJURY
ACTIONS	MAINTAIN SBP TO MAINTAIN NORMAL CONSCIOUS LEVELS (>90mmHg) IF SBP <90mmHg or REDUCED CONSCIOUSNESS: INFUSE BLOOD or FLUID SBP 90-100mmHg: SLOWLY REDUCE RATE OF PHBT or FLUIDS UNTIL SBP 110mmHg SBP >110mmHg: SLOW OR STOP PHBT or FLUIDS SHOULD BE USED WITHIN 1 HOUR OF INJURY, THEN AIM FOR NORMOTENSION
AVOID IN	SEVERE HEAD INJURED PATIENTS, PAEDIATRICS or ELDERLY, AND PREGNANCY

TOURNIQUET (TQ) DOWNGRADE and CONVERSION TO WOUND PACKING DRILLS	
DOWNGRADE CRITERIA	HAEMODYNAMICALLY STABLE, WOUND PACKING RESOURCES and MONITORING AVAILABLE TQ IS NOT NECESSARY or CAN BE APPLIED MORE EFFICIENTLY SHOULD BE CONSIDERED IN PROLONGED CASUALTY CARE SETTINGS >2 HOURS
DOWNGRADE ACTIONS	1. ENSURE APPROPRIATE AND SUFFICIENT ANALGESIA/SEDATION PRIOR TO DOWNGRADE. 2. ASSESS LIMB/WOUND FOR APPROPRIATE INTERVENTION AND PREPARE PACKING KIT. 3. IF TQ IS REQUIRED (AMPUTATION) THEN PLACE NEW TQ 2-3 INCHES ABOVE WOUND ON VIABLE TISSUE (CONSIDER 2nd TQ ON LEGS). 4. ATTEMPT TO PACKAGE WOUND WITH HAEMOSTATIC GAUZE AND STERILE PLAIN GAUZE THEN ECB TO SECURE WOUND PACKING (CONSIDER PROCEDURAL SEDATION) 5. SLOWLY RELEASE INITIAL TQ A QUARTER TURN, REASSESS FOR BLEEDING FOR 30 SECONDS, REPEATING THIS UNTIL TQ IS LOOSE BUT KEEP IN SITU. 6. IF BLEEDING RESTARTS, THEN RETIGHTEN INITIAL TQ AND REASSESS PATIENT. 7. SEEK FURTHER GUIDANCE FROM REACHBACK REGARDING SECOND ATTEMPT BUT IT SHOULD NOT BE CONSIDERED UNTIL DOUBLE THE TIME HAS PASSED SINCE INITIAL TQ APPLICATION AND IT IS FEASIBLE TO DO SO IN PROLONGED CASUALTY CARE SETTING.

ISOTONIC FLUID RESUSCITATION GUIDANCE
IF PHBT IS NOT AVAILABLE/EXHAUSTED and UNABLE TO DO EMERGENCY DONOR PANEL (EDP) CONSIDER 250ml FLUID BOLUSES FOR SHOCKED PATIENTS AND REASSESS FOR RADIAL PULSE OR ADEQUATE SBP AS SEEN ABOVE. MAX DOSE OF ISOTONIC FLUIDS: 20ml/kg

POINTS TO NOTE

1. **IN THE NON-PERMISSIVE/CARE UNDER FIRE (CUF) ENVIRONMENT, IT IS ACCEPTABLE TO PLACE A TQ HIGH AND TIGHT ON THE LIMB BUT THE PATIENT SHOULD BE MOVED TO A MORE PERMISSABLE ENVIRONMENT TO ASSESS THE WOUND PROPERLY IF FEASIBLE AND PLACE TQ AS LOW AS VIABLY POSSIBLE.**
2. **TQ AND WOUND PACKING WILL BE EXTREMELY PAINFUL SO ANALGESIA SHOULD BE CONSIDERED IN ACCORDANCE WITH CONSCIOUS LEVELS AND GUIDELINES.**
3. **TORSO CAVITIES WILL REQUIRE EXTENSIVE PACKING WHICH IS SELDOM FEASIBLE IN THE PREHOSPITAL SETTING DUE TO RESOURCES SO RECOGNITION AND URGENT CONVEYANCE TO DCS IS PARAMOUNT.**
4. **A MAIN INDICATION FOR FLUID RESUSCITATION IN PENETRATING THORACIC TRAUMA IS MENTAL STATUS. REDUCED CONSCIOUSNESS SHOULD PROMPT PHBT PROTOCOLS IF FEASIBLE.**
5. **A PELVIC BINDER SHOULD BE PLACED IF THE MECHANISM INDICATES AND THERE IS CONCERN FOR ANY INTERNAL HAEMORRHAGE (SHOCKED PATIENT, PAIN, PERINEAL BRUISING, URETHRAL BLEEDING, ETC.)**
6. **A RADIAL PULSE IS SUGGESTIVE OF ADEQUATE PERIPHERAL BLOOD PRESSURE AND CAN BE USED AS A INDICATOR OF FLUID IN THE ABSENCE OF MONITORING EQUIPMENT**
7. **1g TXA SHOULD BE ADMINISTERED INTRAVENOUSLY OVER 10 MINUTES AND WITHIN 3 HOURS OF INJURY. IT CAN BE USED FOR PATIENTS WITH SUSPECTED HEAD INJURIES.**
8. **CALCIUM CHLORIDE CAN CAUSE PERIPHERAL VASCULAR IRRITATION SO SHOULD BE FLUSHED WITH 20ml SALINE.**
9. **TQ DOWNGRADE IDEALLY SHOULD OCCUR IN UNDER 2 HOURS AND UNABLE TO CONVEY TO DEFINITIVE CARE IN A TIMELY MANNER.**
10. **ISOTONIC FLUIDS ARE A BETTER CHOICE THAN NOTHING.**

Guideline written in accordance with SFM Clinical Standard Operating Guideline 005 'Traumatic Haemorrhage' (2022) and MERT SOP 01 'Blood Product and Fluid Resuscitation in Trauma V3.0' (2022).

References

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